

Guidelines: Fire and Gas Emergency Monitoring System Operational Matrix

NOTES:		OUTPUT																
		Gas Shutdown Lab	Gas Shutdown Lab	Gas Shutdown Lab	Gas Shutdown Lab	TGO Building	TGO Lab	TGO Bunker	TGO Bunker	Fire Department	Emergency Ctr	Emergency Ctr	Emergency Ctr	Emergency Ctr	Emergency Ctr	Fire Alarm	Fire Department	
1. Gas shutdown is for area affected only. For example, if treatment failure is for a gas cabinet, then only shutdown gas flow to that cabinet.																		
2. EMCS (Emergency Management Control System) Level 1 is for same or next workday response.																		
3. EMCS Level 2 is for immediate response; however, local responders may adjust their response time depending on the type of alarm and known information.																		
4. Control systems shall use GE, Allen Bradley, or Seimens type PLC.																		
5. TGO strobes shall be "blue" in color.																		
6. Time Delay for Excess Flow is 10 sec for 0-100ft, 20 sec for 101-200ft, 30 sec for 201-300ft, and 40 sec for 301ft or greater length of piping.																		
7. Gas detection alarm may be delayed up to 45 sec for PELs, but must be instantaneous for IDLHs.																		
8. Emergency Power Off [EPO] switches must have green-off and red-on light indicators.																		
9. Local users are designated by lower case "x" and upper case "X" with lower case being the research user and upper case being the user's unit safety contact.																		
10. "y" designation indicates additional outputs to the Fire Alarm outputs if the TGO Alarm is activated.																		
11. If a bunker is separate from a building, during a bunker evacuation, only the bunker will be evacuated.																		
INPUT	Seismic switch	x	x	x	x						x				x			
	Gas detection (Lab room air)	x	x			x	x			x			x		x	x	x	
	Gas detection (Lab exhausted area)	x	x				x						x	x				
	Gas detection (Inside storage, room air)								x	x			x		x	x	x	
	Gas detection (Bunker flammable, room air)			x					x				x	x		x	x	
	Gas detection (Bunker flammable, exhausted air)			x					x				x	x				
	Gas detection (Bunker toxic, room air)				x				x	x			x	x		x	x	
	Gas detection (Bunker toxic, exhausted area)				x				x				x	x				
	Gas detection, low level										x							
	Gas detection fault										x			x				
	Emergency power off [EPO] (Remote, if used)	x	x	x	x								x	x				
	Emergency power off [EPO] (Lab)	x	x										x	x				
	Emergency power off [EPO] (Bunker)			x	x								x	x				
	Failure of emergency power [Time delay = 5 sec]	x	x	x	x							x			x			
	Failure of treatment system	x	x	x	x							x		x				
	Excess flow [Time delay]	x										x			x			
	Excess flow [Time delay] (Bunker, flammable)			x								x			x			
	Excess flow [Time delay] (Bunker, toxic)				x							x			x			
	Fire/TGO alarm activation building (smoke, pull, water flow)	x	x	x	x		y					x				x	x	x
	Fire/TGO alarm activation bunker (smoke, pull, water flow)			x	x				y			x				x	x	x